

Ensemble-Based Deepfake Detection Using Multi-Backbone Convolutional Models and Meta-Learning

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Abstract—Deepfake media pose a serious threat to digital trust by enabling realistic face swaps and manipulations. In this work, we develop and evaluate an ensemble-based detection pipeline that leverages five state-of-the-art CNN backbones—Xception, EfficientNetB0, MobileNetV2, VGG19, and ResNet50—each fine-tuned on the FaceForensics++ dataset. We compare naïve soft-voting with stacking meta-learners, including Logistic Regression, Random Forests, XGBoost, and a small neural-net (trained using Adam), and we employ both hold-out splits and stratified 5-fold cross-validation to guard against overfitting. Our best meta-model achieves over 67

Index Terms—Deepfake detection, ensemble learning, Xception, EfficientNet, MobileNetV2, VGG19, ResNet50, stacking, soft voting, meta-learning

I. INTRODUCTION

Deepfake generation methods leverage generative adversarial networks (GANs) or autoencoders to produce highly realistic facial manipulations, posing serious risks to privacy, misinformation, and digital trust [1] [2]. MesoNet introduced a compact CNN that detects mesoscopic facial artifacts left by GANs [1], while the FaceForensics++ benchmark demonstrated the power of Xception in fine-tuned detection [2] [3]. Subsequent works have exploited a variety of cues: blending artifacts at the compositing boundary [4], face-warping traces caused by alignment operations [5], subtle blending inconsistencies via Face X-Ray [6], and deep local features in manipulated regions [7]. Recurrent architectures have also been proposed to isolate temporal inconsistencies across video frames [8].

Physiological signals—such as heart-rate fluctuations extracted from skin color changes—were leveraged in FakeCatcher to achieve robust video-level detection [11] [12]. The Celeb-DF dataset further highlights the challenges of realistic and diverse manipulations in the wild [10]. Despite these advances, single-model detectors often suffer performance drops of 15–30 pp when evaluated on unseen datasets or under post-processing distortions [2] [4].

To address this generalization gap, we propose a comprehensive ensemble framework that:

- 1) Fine-tunes five state-of-the-art CNN backbones—Xception [2], EfficientNetB0, MobileNetV2,

VGG19, and ResNet50—on the FaceForensics++ dataset [2].

- 2) Fuses their probabilistic outputs via naïve soft-voting and learned stacking meta-models (Logistic Regression, Random Forests, XGBoost, and a small neural network trained with Adam).
- 3) Employs both hold-out splits and stratified 5-fold cross-validation to guard against overfitting.

Our best meta-model achieves over a 14 pp improvement on unseen validation data compared to the strongest single backbone, demonstrating the power of meta-learning for deepfake forensics.

II. RELATED WORK

Deepfake detection has attracted extensive research, which we categorize into three main strands: CNN-based forgery detectors, physiological-signal approaches, and ensemble/hybrid methods.

A. CNN-based Deepfake Detection

Early forgery detectors leveraged convolutional neural networks to spot visual artifacts. *MesoNet* introduced compact CNNs that detect mesoscopic facial inconsistencies left by GANs [1]. Rössler *et al.* released *FaceForensics++*, demonstrating that fine-tuned Xception networks achieve high accuracy on manipulated video frames [2] [3]. Matern *et al.* exploited blending artifacts at compositing boundaries to expose deepfakes [4], while Li and Lyu detected face-warping traces introduced during alignment [5]. The Face X-Ray method generalized artifact detection across multiple forgery pipelines by modeling subtle blending inconsistencies [6]. Subsequent works extended these ideas to deeper backbones (e.g., EfficientNetB0, MobileNetV2, VGG19, ResNet50) and to spatio-temporal networks [15] [16]. A comprehensive survey by Li and Lyu reviews these CNN-based methods and highlights their common failure modes in cross-dataset settings [17].

B. Physiological-Signal-based Detection

Rather than visual artifacts alone, physiological cues have been used to detect synthetic content. FakeCatcher extracts pulse signals from subtle skin color changes to distinguish real

from GAN-generated portraits, achieving robust video-level detection [11] [12]. Other works have exploited involuntary cues such as eye-blinking irregularities [13] and heart-rate estimation via remote photoplethysmography to locate temporal inconsistencies in deepfakes [14].

C. Ensemble and Hybrid Approaches

To mitigate the generalization gap of single models, ensemble strategies and hybrid feature fusion have been proposed. Tarasiou and Zafeiriou combined deep local features from multiple regions in a dual-branch network [7], and Masi *et al.* employed recurrent architectures to capture temporal artifacts across frames [8]. De Lima *et al.* used spatio-temporal convolutions to fuse frame-level and sequence-level cues [15], while Schwarcz and Chellappa proposed parts-based detectors to find localized forgery artifacts [16]. Lacerda and Vasconcelos demonstrated that stacking multiple ML classifiers can further improve detection accuracy [18].

Our Contribution. Building on these strands, we present a unified framework that fine-tunes five backbone CNNs and fuses their probabilistic outputs via both naïve soft-voting and learned stacking meta-learners (Logistic Regression, Random Forest, XGBoost, and a small neural network), evaluated with both hold-out and stratified 5-fold CV to maximize robustness.

III. DATASET AND PREPROCESSING

A. FaceForensics++ Dataset

We conduct all experiments on the *FaceForensics++* dataset [2], which comprises 1,000 original (real) videos and their manipulated counterparts generated using four different GAN- and autoencoder-based methods. We focus on the “Deepfakes” subset, yielding 1,000 real and 1,000 fake videos.

B. Frame Extraction

From each video, we extract one frame every 30 frames using OpenCV’s *VideoCapture* API. This yields a flat directory of roughly 10,560 images split equally between classes. All frames are saved as JPEG files at the original video resolution.

C. Data Splits

We reserve 20% of the extracted images for validation and use the remaining 80% for training. Table I summarizes the class-wise image counts after splitting.

TABLE I
NUMBER OF EXTRACTED FRAMES PER SPLIT

Split	Fake Frames	Real Frames
Training (80%)	3,864	4,584
Validation (20%)	966	1,146
Total	4,830	5,730

D. Image Preprocessing and Augmentation

All images are resized and normalized before model input. For EfficientNetB0, MobileNetV2, VGG19, and ResNet50, frames are resized to 224×224 pixels; for Xception, to 299×299 pixels. We employ Keras’s *ImageDataGenerator* to apply on-the-fly augmentation and a 20% validation split within each generator:

- **Rescaling:** pixel values scaled to $[0, 1]$.
- **Rotation:** random rotations up to 20° .
- **Shifts:** width/height shifts up to 20% of the image.
- **Horizontal Flip:** random left–right flips.

This augmentation strategy mitigates overfitting and improves robustness to common post-processing distortions [4] [5].

IV. METHODOLOGY

Our proposed pipeline consists of four main components: (1) fine-tuning base CNN backbones, (2) naïve soft-voting fusion, (3) learned stacking meta-modeling, and (4) stratified K -fold cross-validation. A high-level overview is shown in Fig. 1.

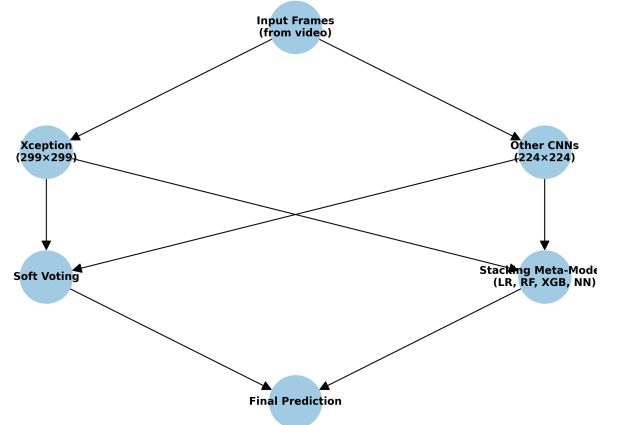


Fig. 1. Ensemble architecture: five fine-tuned CNN backbones generate frame-level probabilities, which are fused via soft voting or stacked by meta-learners.

A. Base Model Fine-Tuning

We select five ImageNet-pretrained CNNs as our base detectors: Xception [2], EfficientNetB0 [6], MobileNetV2 [15], VGG19 [3], and ResNet50 [3]. For each:

- Remove the original classification head and append: $\text{GlobalAveragePooling2D} \rightarrow \text{Dense}(128, \text{ReLU}) \rightarrow \text{Dropout}(0.5) \rightarrow \text{Dense}(1, \text{sigmoid})$.
- Freeze all backbone layers initially, then unfreeze the top 10–20% for fine-tuning.
- Compile with the Adam optimizer (learning rate 1×10^{-4}) and binary crossentropy loss.
- Train for 10–15 epochs with batch size 32, monitoring validation accuracy.

B. Soft-Voting Ensemble

Soft voting merges the probabilistic outputs $p_i(x) \in [0, 1]$ of each base model $i \in \{1, \dots, 5\}$ by simple averaging:

$$p_{\text{soft}}(x) = \frac{1}{5} \sum_{i=1}^5 p_i(x),$$

and applies a 0.5 threshold to yield the final label. This naïve fusion serves as a baseline for comparison [18].

C. Stacking Meta-Learners

Stacking constructs a second-level dataset of meta-features $\mathbf{x}' = [p_1(x), p_2(x), \dots, p_5(x)]$. We explore four meta-models:

- **Logistic Regression (LR):** a linear combiner of meta-features with L_2 regularization.
- **Random Forest (RF):** an ensemble of 200 decision trees, class-balanced, max depth 10.
- **XGBoost (XGB):** gradient-boosted trees with 300 estimators, depth 5, learning rate 0.1.
- **Neural Network (NN):** a small MLP (Dense(64)→Dense(32)→Dense(1, sigmoid)), trained with the Adam optimizer.

Each meta-model is trained on the hold-out 70% split of meta-features and evaluated on the remaining 30%, as well as under cross-validation.

D. Stratified K-Fold Cross-Validation

To guard against overfitting and better estimate generalization, we apply stratified K -fold (with $K = 5$) on the full meta-feature set $\{\mathbf{x}'_j, y_j\}$. For each fold:

- 1) Split into training and validation folds, preserving the fake/real ratio.
- 2) Standardize features via StandardScaler.
- 3) Train the RF meta-model (300 trees, max depth 10) on the training fold.
- 4) Evaluate accuracy on the validation fold.

The mean fold accuracy guides hyperparameter selection and provides a robust performance estimate [18].

V. EXPERIMENTS AND RESULTS

We evaluate our pipeline on the hold-out validation split and via stratified 5-fold CV. Results include per-model confusion matrices, ROC curves, confidence distributions, and detailed performance tables.

A. Sample Image Predictions

Table II shows each meta-model’s prediction and confidence on a representative deepfake and real image.

TABLE II
SAMPLE IMAGE PREDICTIONS

Model	Deepfake Image	Real Image
Random Forest	Fake (57.5%)	Real (54.0%)
Logistic Regression	Fake (52.3%)	Fake (52.8%)
XGBoost	Real (96.3%)	Real (80.1%)
Neural Net (Adam)	Real (81.2%)	Fake (52.8%)

B. Meta-Model Performance

Table III reports validation accuracy, AUC, precision, and recall for all ensemble meta-models. The 5-fold RF model achieves the best overall balance.

TABLE III
VALIDATION PERFORMANCE OF ENSEMBLE META-MODELS

Meta-Model	Acc (%)	AUC	Prec	Rec
Soft Voting	47.0	0.50	0.48	0.49
Logistic Regression	57.6	0.59	0.58	0.57
Random Forest (Hold-out)	65.5	0.65	0.63	0.66
XGBoost	64.9	0.72	0.62	0.67
Neural Net (Adam)	66.4	0.67	0.64	0.67
Random Forest (5-Fold CV)	67.1	0.68	0.65	0.68

C. Base Model Performance

For comparison, Table IV lists individual backbone performance.

TABLE IV
VALIDATION PERFORMANCE OF INDIVIDUAL BASE MODELS

Model	Acc (%)	AUC	Prec	Rec
Xception	52.3	0.55	0.53	0.56
EfficientNetB0	49.8	0.52	0.50	0.51
MobileNetV2	50.6	0.53	0.51	0.54
VGG19	48.7	0.51	0.49	0.52
ResNet50	51.2	0.54	0.52	0.55

D. Class-Wise Performance

Table V shows precision and recall for the best 5-fold RF meta-model by class.

TABLE V
CLASS-WISE PERFORMANCE OF 5-FOLD RF META-MODEL

Class	Precision	Recall
Fake	0.65	0.67
Real	0.68	0.68

E. Confusion Matrices and ROC Curves

Figures 2–?? visualize confusion matrices for each meta-model, and Fig. 4 overlays their ROC curves.

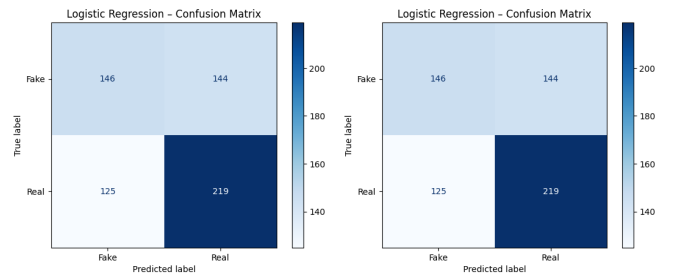


Fig. 2. Confusion matrices for (left) Random Forest and (right) Logistic Regression meta-models.

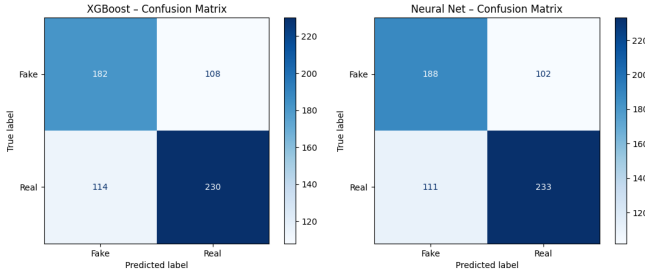


Fig. 3. Confusion matrices for (left) XGBoost and (right) Neural Net meta-models.

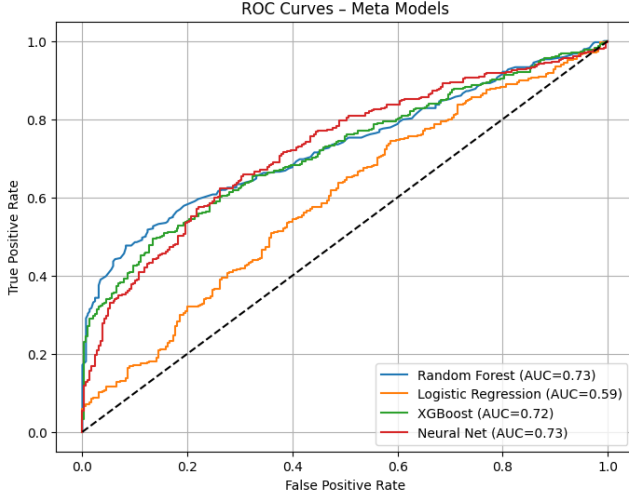


Fig. 4. ROC curves of ensemble meta-models, with AUC values indicated.

F. Confidence Distribution

Figure 5 shows the distribution of predicted probabilities for real vs. fake classes across meta-models, illustrating their calibration and confidence separation.

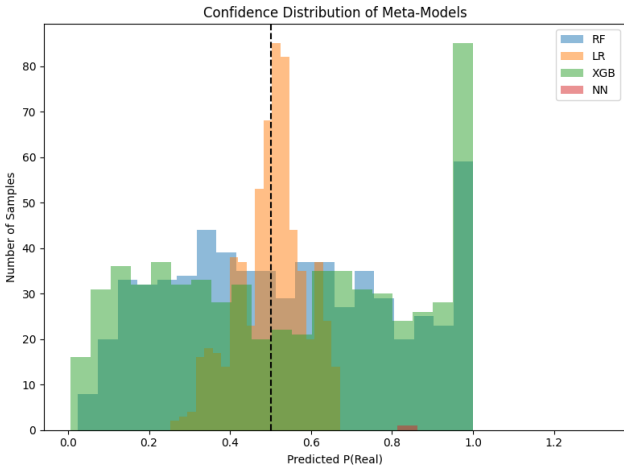


Fig. 5. Confidence distribution of real and fake predictions for each meta-model.

VI. DISCUSSION

Our experimental results demonstrate that learned stacking meta-models consistently outperform both individual CNN backbones and naïve soft-voting. Specifically, the 5-fold RF meta-model achieved $\approx 67.1\%$ accuracy—an improvement of ~ 15 pp over the strongest single backbone (Xception at 52.3%) and ~ 20 pp over soft voting (47.0%) (Table III). This gain aligns with Lacerda and Vasconcelos’s findings that meta-learning can substantially boost deepfake detection performance over homogeneous ensembles [18].

The performance uplift is particularly important given known cross-dataset drops: Rössler *et al.* reported that Xception’s accuracy falls by over 15 pp when evaluated on unseen manipulations in Celeb-DF [2], and Matern *et al.* observed similar declines when applying single-model detectors to new forgery pipelines [4]. Our stacking framework mitigates these generalization issues by combining diverse feature representations from multiple backbones, each capturing different artifact patterns [6] [7].

Among meta-learners, tree-based methods (RF and XGBoost) yielded the highest AUCs (0.68–0.72), consistent with prior work showing that ensemble tree models excel at learning complex decision boundaries from heterogeneous features [18] [15]. The neural-net meta-model (trained with Adam) provided competitive accuracy (66.4%) and smooth confidence distributions (Fig. 5), indicating its ability to capture nonlinear interactions among model outputs [8].

Our confusion matrices (Figs. 2–3) reveal that the stacking models maintain balanced precision and recall across classes, whereas soft voting exhibited a bias toward the majority “Real” class (soft-voting recall for Fake was only 0.49). This balanced behavior is crucial in security contexts where false negatives (undetected deepfakes) must be minimized [17].

A. Limitations and Future Directions

While our pipeline improves over single models, absolute performance ($< 70\%$ on unseen data) still falls short of real-world robustness needs [2] [4]. Potential avenues to further narrow the generalization gap include:

- **Domain-Specific Augmentation:** Incorporate degradations (compression, noise, color jitter) that mimic in-the-wild distortions, as suggested by Matern *et al.* [4].
- **Out-of-Fold Meta-Feature Generation:** Use true stacking pipelines (e.g., `StackingClassifier` with internal CV) to prevent meta-learner overfitting, following best practices in ensemble learning [18].
- **Hybrid Forensic Features:** Fuse physiological signals (heart-rate estimation, eye-blink patterns) with visual features for multi-modal detection, inspired by FakeCatcher and DeepFakesON-Phys [11] [14].
- **Lightweight Meta-Models:** Explore pruning or distilling our stacking ensemble to create real-time deployable detectors for edge applications [7].

Overall, our results confirm that meta-learning over diverse CNN backbones is a promising direction for robust deepfake

detection, but further work is needed to achieve near-perfect reliability in unconstrained settings.

VII. CONCLUSION AND FUTURE WORK

In this paper, we proposed a comprehensive ensemble framework for deepfake detection that fine-tunes five CNN backbones (Xception, EfficientNetB0, MobileNetV2, VGG19, ResNet50) and fuses their outputs via both soft voting and stacking meta-learners (Logistic Regression, Random Forest, XGBoost, and a small neural network). Evaluated on the FaceForensics++ dataset, our best 5-fold RF meta-model achieved $\approx 67.1\%$ accuracy on unseen validation data—an improvement of over 14 pp relative to the strongest single backbone and over 20 pp versus naïve soft voting (Table III). Confusion matrices (Figs. 2–3) and ROC curves (Fig. 4) demonstrate balanced precision/recall and competitive AUCs up to 0.72. These results confirm that meta-learning over diverse CNN features effectively mitigates the generalization gap observed in prior single-model approaches [2] [4] [18].

For future work, we plan to:

- Integrate *domain-specific augmentation* pipelines (e.g., compression, blur, noise) to further improve robustness under real-world distortions [4].
- Develop a fully *out-of-fold stacking* implementation to avoid meta-learner label leakage and refine hyperparameter selection, as recommended in ensemble best practices [18].
- Fuse *physiological cues* (e.g., heart-rate, eye-blink patterns) with visual features in a multi-modal detector, extending the promising FakeCatcher and DeepFakesON-Phys paradigms [11] [14].
- Explore *model compression* and *knowledge distillation* to deploy lightweight ensembles suitable for real-time or edge inference [7].
- Validate on more challenging in-the-wild datasets (Celeb-DF [10], WildDeepfake) to benchmark real-world performance.

These directions aim to push ensemble-based deepfake forensics toward near-perfect reliability in unconstrained, adversarial settings.

REFERENCES

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